

Different Types Of Rolling Mills

Rolling is the most rapid method of forming metal into desired shapes by plastic deformation through compressive stresses using two or more than two rolls. There are different types of rolling mills which are described as under.

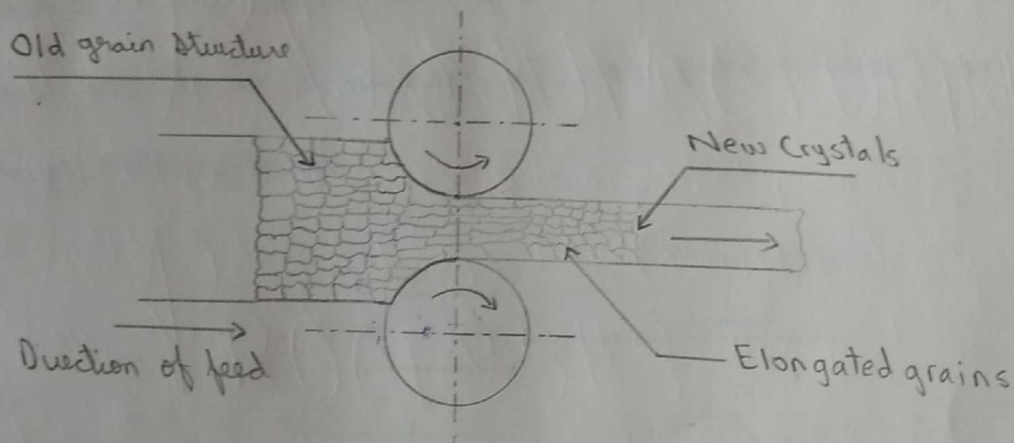


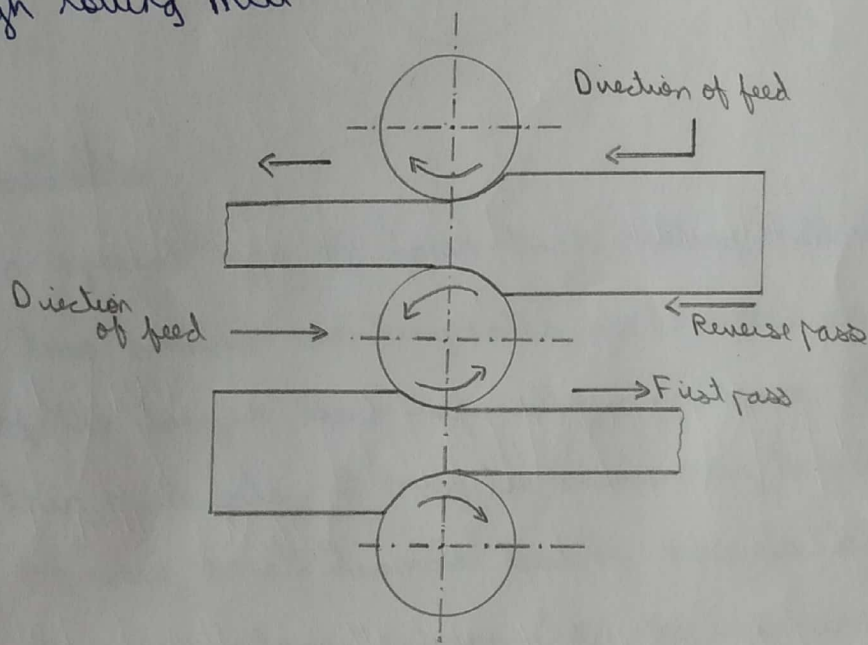
Fig A.1 Grain refinement in hot rolling process

Two-High Rolling Mill

A two high rolling mill has two horizontal rolls revolving at the same speed but in opposite direction. The rolls are supported on bearings housed in sturdy upright side frame called stands. The space between the rolls can be adjusted by raising or lowering the upper roll. Their direction of rotation is fixed and cannot be reversed. The reduction in the thickness of work is achieved by feeding from one direction only.

Three-High Rolling Mills

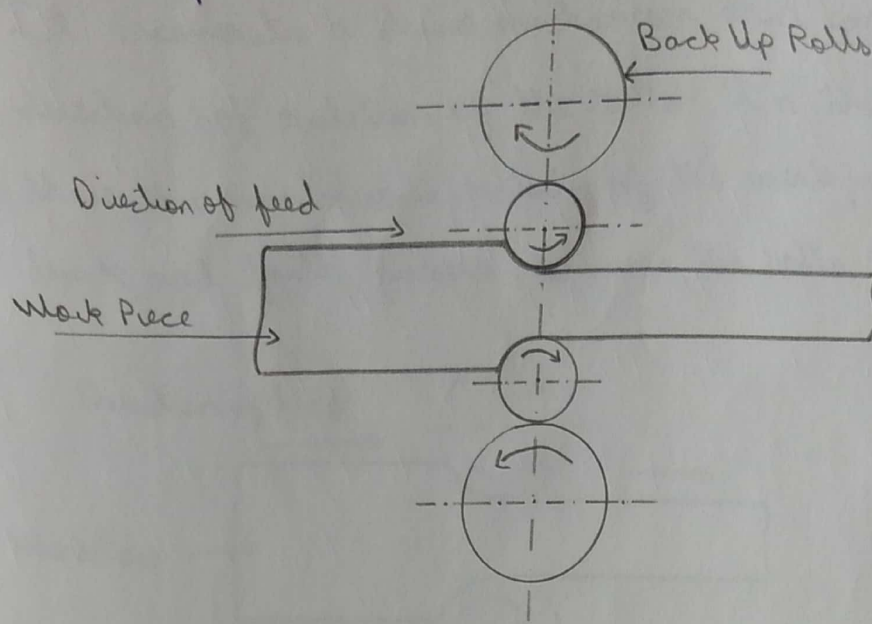
It consists of three parallel rolls, arranged one above the other as shown. The directions of rotation of the upper and lower rolls are the same but the intermediate roll rotates in a direction opposite to both ~~sides~~ of these. This type of rolling mill is used for rolling of two continuous passes in a rolling sequence without reversing the dimes. This results in a higher rate of production than the two high rolling mill.



Four-High Rolling Mill

It is essentially a two-high rolling mill, but with small sized rolls. Practically, it consists of four horizontal rolls, the two middle rolls are smaller in size than the top and bottom rolls as shown. The smaller size rolls are known as working rolls which concentrate the total rolling pressure over the workpiece. The larger diameter rolls are called back-up rolls and their main function is to prevent the deflections of the smaller rolls, which otherwise would result in thickening of rolled plates or sheets.
 ... is a compromise, it is

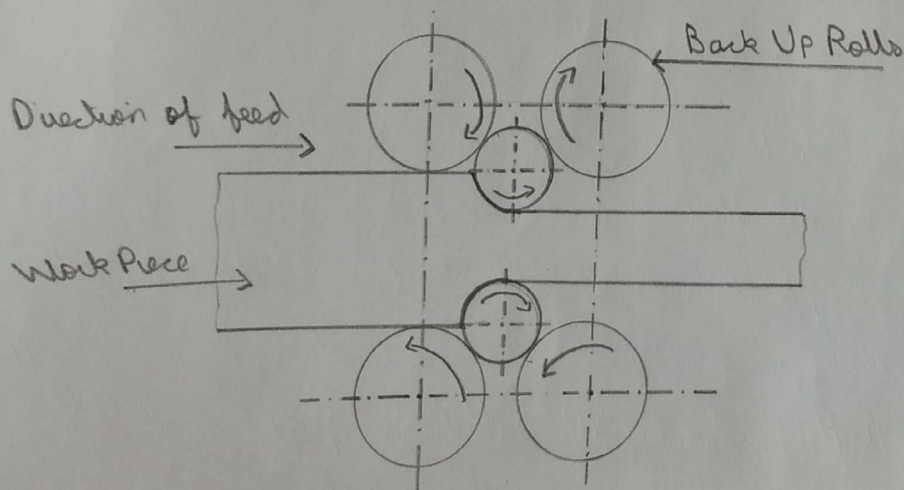
at the centre. The common products of these mills are hot or cold rolled plates and sheets.



Cluster Mills

It is a special type of four-high rolling mill in which each of the two smaller working rolls are backed up by two or more of the larger back-up rolls as shown. For rolling hard thin materials, it may be necessary to employ work rolls of very small diameter but of considerable length.

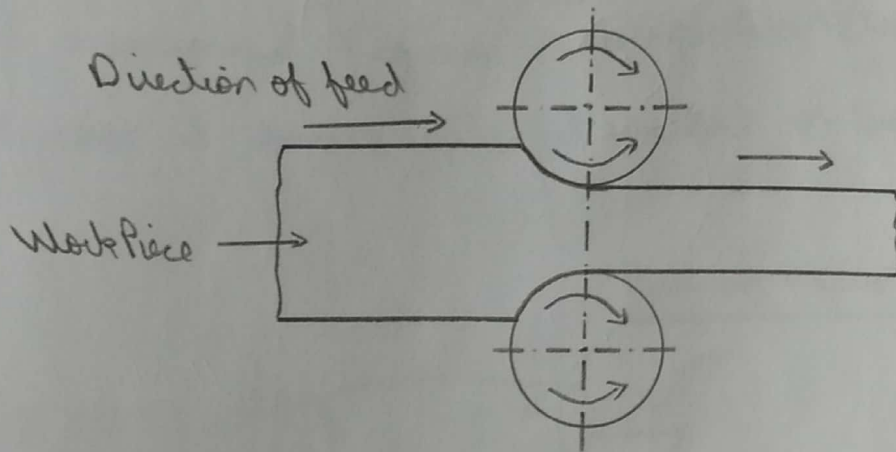
In such cases adequate support of the working rolls can be obtained by using a cluster mill. This type of mill is generally used for cold rolling work.



is created. While it is run up to a temperature, it is

Two-High Reverse Rolling

It incorporates a drive mechanism that can reverse the direction of rotation of the rolls. In this mechanism, there is a continuous rolling of the workpiece through back and forth passes between the rolls.

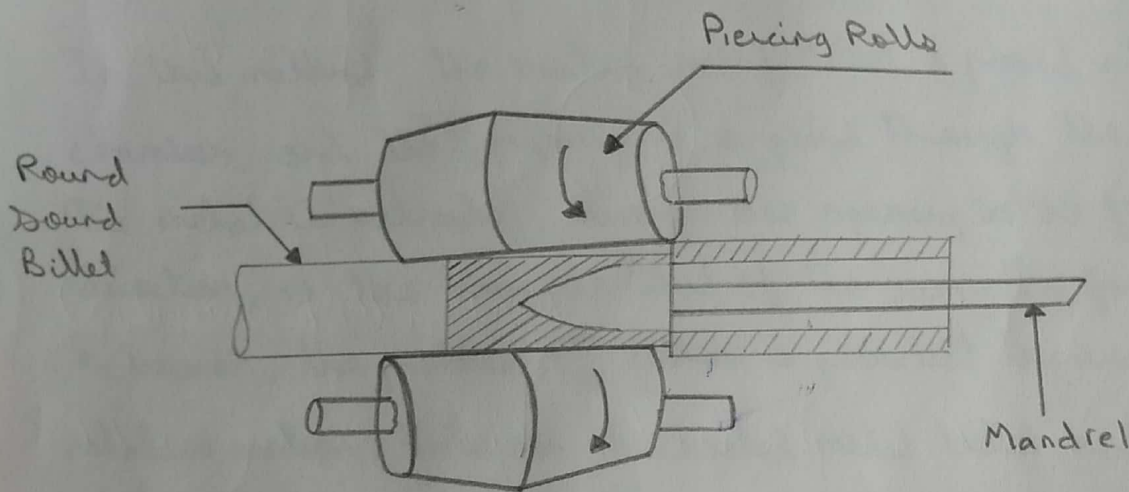


Continuous Rolling Mill

It consists of a number of non reversing two-high rolling mills arranged one after the other, so that the material can be passed through all of them in sequence. It is a suitable for mass production work only, because for smaller quantities quick changes of set up will be required and they will consume lot of time and labor.

2. How are seamless tubes manufactured by rolling process?

Hot piercing or Seamless tubing is a rolling process used to manufacture seamless tubes. It is used for making thin walled round objects. Seamless tubes is popular and economical process in comparison to machining because it saves material wasted in boring of parts.



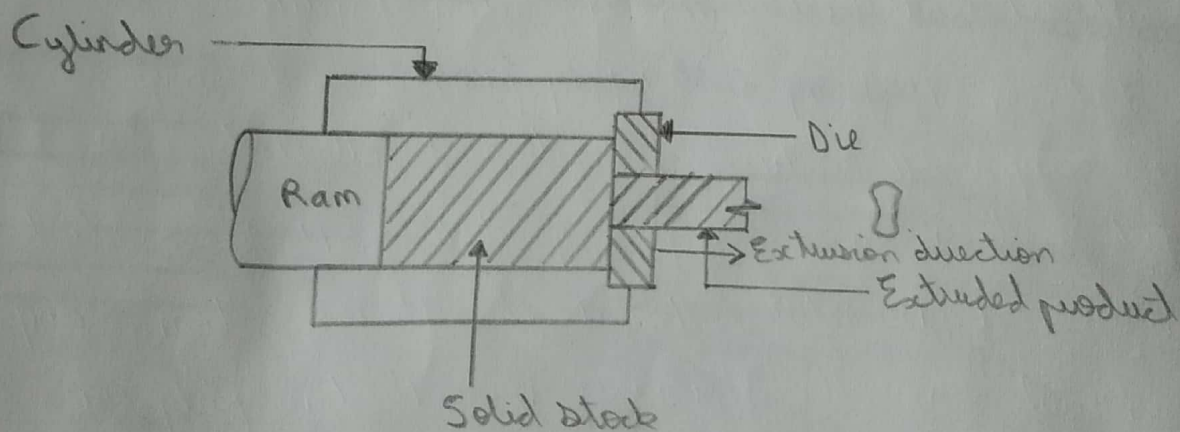
Hot piercing includes rotary piercing to obtain formed tubes by piercing a pointed mandrel through a billet in a specially designed rolling mill. The rotary piercing can be performed either on a two-high rolling mill or on a three-high rolling mill. In the former, the two rolls are set at an angle to each other. The billet under the rolls is deformed and a cavity formation is initiated at the center due to tensile stressing. The carefully profiled shape of the mandrel assists and controls the formation of cavity. The squeezing and bulging of the billet open up a seam in its center. The ~~path~~ pass makes a rather thick walled tube which is again passed over plug and through grooved rolls in a two-high roll mill where the thickness is decreased and the length is increased. While it is still up to a temperature, it is

passed on to a reeling machine which has two rolls similar to the piercing rolls, but with flat surfaces. If more accuracy and better finish are desired, the run through sizing dies on rolls. After cooling, the tubes are used in a pickling bath of dilute sulphuric acid to remove the scale.

3 What are different types of extrusion processes employed?

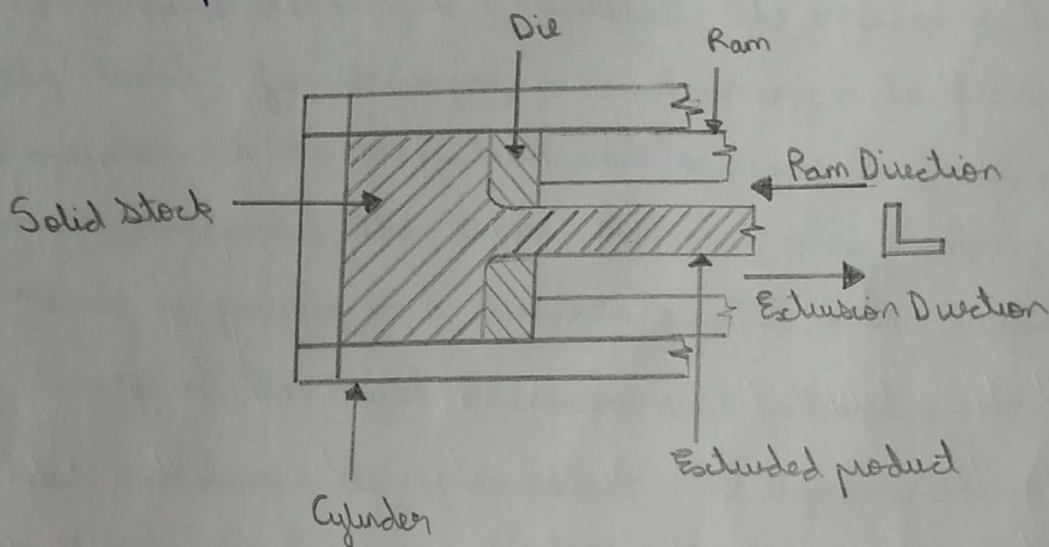
Direct or Forward Hot Extrusion

In this method, the heated metal billet is placed into the die chamber and the pressure is applied through the ram. The metal is extruded through die opening in the forward direction, i.e. the same as that of the ram. In forward extrusion, the problem of friction is prevalent because of the relative motion between the heated metal billet and the cylinder walls. Lubricants are used to reduce the friction. At lower temperatures, a mixture of oil and graphite is generally used. The problem of lubrication gets compounded at higher operating temperatures. Molten glass is generally used for extruding steels.



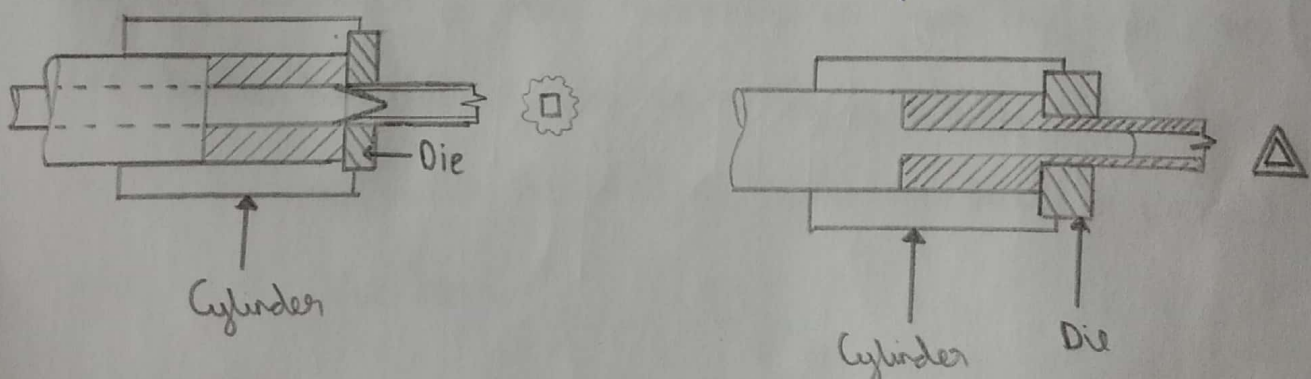
Indirect or Backward Hot Extrusion

In indirect hot extrusion, the billet remains stationary while the die moves into the billet by the hollow ram (or punch), through which the backward extrusion takes place. Since there is no friction between the billet and the container wall, therefore less force is required by this method. However, this process is not widely used because of the difficulty occurred in providing support for the extruded part.



Tube Extrusion

This process is an extension of direct extrusion process where additional mandrel is needed to restrict flow of metal for production of seamless tubes. Aluminium based toothpaste and medicated tubes are produced using this process.



Q What do you understand by

- 1) Upsetting
- 2) Folding
- 3) Press Forging
- 4) Closed die forging

Upsetting

Upsetting is also known as jumping operation which is carried out to increase the thickness (or diameter) of a bar and to reduce its length. Generally the increase in thickness is only local, for example, when forming a bolt head. This operation is an operation just opposite to drawing and involves increasing the cross-sectional area usually by hammering or pressing in a direction parallel to the longitudinal axis. The length of the rod decreases and following the path of least resistance it spreads out. The required shape is given the rod by spreading it between two dies. Only that portion of the bar which is to be upset is heated in water so that upset will form only on the hot portion of the bar. In one method of upsetting, the bar is held in the tong and supported vertically on the anvil. The top edge of the bar is then hammered to form the upset on the bottom hot end of the bar. For upsetting, the blow of the hammer must be in line with the bar to prevent bending of the bar.

Folding or Bending

Bending is very commonly used forging operation in forging shop to give a turn of to a metal rod or plate. It is accompanied by spreading of the metal in the inside of the bend and narrowing at outside. The simplest method of bending a piece of metal in hand forging is to support it on the anvil and to strike its free end with a hammer. When bent, the metal of the workpiece thins out round bend causing weakness. This can be overcome by upsetting the bar prior to bending.

Press Forging

In press forging, the metal is shaped not by means of a series of blows as in drop forging but by means of single continuous squeezing action. This squeezing is obtained by means of presses. Owing to the continuous action of the press, the material gets uniformly deformed throughout its entire depth. The impressions obtained in press forging are clear compared to that of the likely panned impressions which are likely in the drop forged components.

The press is generally of vertical type and the squeezing action is carried completely to the centre of part being pressed.

Closed-die forging

In closed die forging, the workpiece is completely surrounded by dies, while in impression die forging, any excess metal in the die cavity is formed into a flash. Since no flash can be formed in closed die forging, proper control of the volume of the material is essential to obtain a forging of desired dimensions (and to avoid generating extreme pressures in the dies from overfilling).

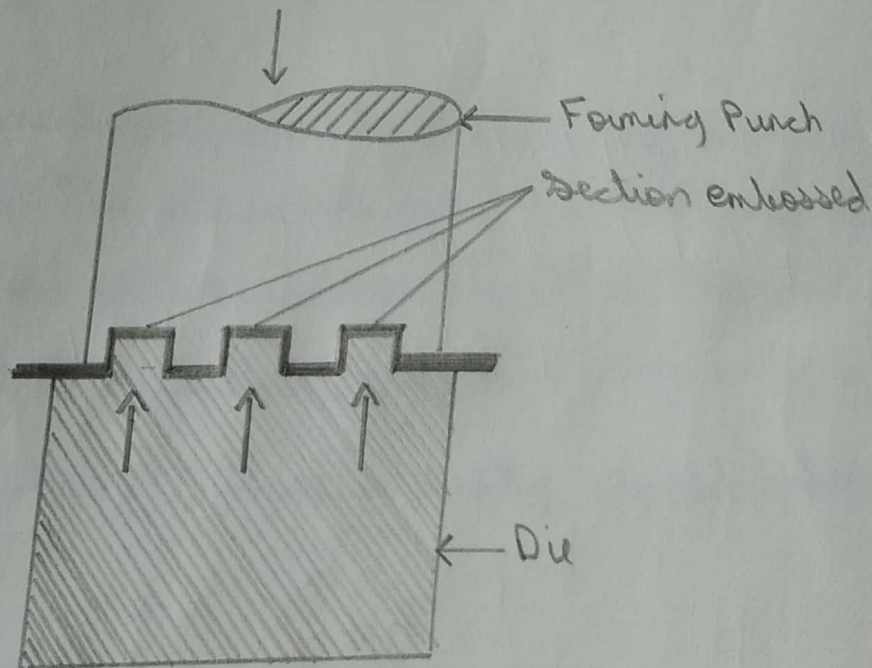
One approach to getting the right amount of metal for the die cavity and reduce forging time is the use of roll-formed shapes or extruded perfor shapes.

5.2 Short notes on

- 1) Coining
- 2) Swaging
- 3) Stamping

Coining

It is basically a cold working operation, which is performed in dies where the metal blank is confined and its lateral flow is restricted. It is mainly used for production of important articles such as medals, coins etc. which possess shallow configurations on their surfaces. The operation involves placing a metal slug in the die and applying heavy pressure by the ~~low~~ punch ~~and the die~~. The metal flows plastically and is squeezed to the shape between punch and the die. The process, on account of the very high pressures required, can be employed only for soft metals with high plasticity.

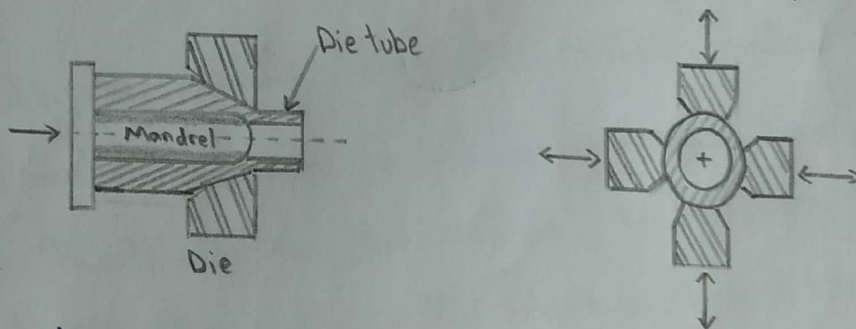


Rotary Forging Or Swaging

This process can be used to reduce diameter of round bars or tubings. The other applications include fabrication of stepped and tapered shafts, pipes with forged out end etc.

The process can be done both in hot and cold state. However cold working is preferable because of the greater use of hardening and better surface finish obtained.

In swaging, also known as rotary swaging or radial forging, a solid rod or tube is reduced in diameter by the reciprocating radial movement of two or four dies.

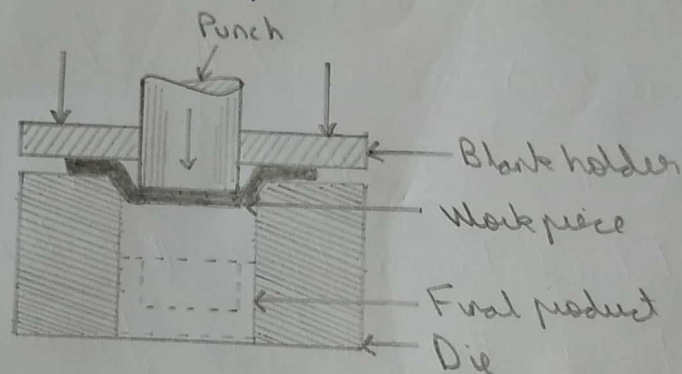


The die movements are generally obtained by means of a set of rollers in cage. The internal diameter and thickness of the tubes can be controlled with or without mandrels. Mandrels can also be made with longitudinal groove (similar in appearance to a splined shaft); thus internally shaped tubes can be swaged.

The swaging process is usually limited to workpiece diameters of about 50 mm, although special machinery has been built up to swage gun barrels of large diameters.

Stamping

In sheet metal, drawing is a process of forming flat sheet metal into hollow shapes by means of a punch that cause the metal to flow into the die cavity. If the depth is one or more times the diameter, the process is called deep drawing. The forming of shallow shapes is sometimes referred to as stamping.



In deep drawing, also called "cup or radial drawing", a parallel, a parallel walled cup is created from a flat blank. The blank may be circular, rectangular, or of a more complex outline. The blank is drawn into the die cavity by action of the punch. Deformation is restricted to the flange and draw radius. No deformation occurs under the bottom of the punch. As punch forms the cup, the amount of material in flange decreases.